

A study indicates that endurance exercise performance is related to type 1 skeletal muscle fiber characteristics that allow for efficient performance of aerobic respiration to generate ATP for extended periods of time. Given this, which of the following characteristics is (are) most likely to be associated with type 1 skeletal muscle fibers?

- I. High mitochondrial density
- II. High numbers of capillaries around the muscle fiber
- III. High myoglobin concentration in the muscle fiber

- ☐ A. I only [2%]
- ☐ B. I and II only [17%]
- ☐ C. III only [2%]
- ☒ D. I, II, and III [77%]

Correct

76% answered correctly

Explanation:

General characteristics of typical skeletal muscle fiber types			
Fiber type	Slow oxidative (type 1)	Fast oxidative-glycolytic (type 2A)	Fast glycolytic (type 2X)
Speed of contraction	Slow	Medium	Fast
Best use	Endurance activity (eg, long-distance running, posture)	Moderate endurance activity (eg, medium-distance running)	Explosive movements (eg, weight lifting, jumping)
Resistance to fatigue	Fatigue resistant	Intermediate susceptibility to fatigue	Easily fatigable
Primary source of ATP	Aerobic respiration	Anaerobic glycolysis & aerobic respiration	Anaerobic glycolysis
Mitochondria	Plentiful	Plentiful to moderately plentiful	Few
Capillaries	Plentiful	Plentiful to moderately plentiful	Few
Myoglobin	Plentiful	Plentiful to moderately plentiful	Few
Appearance	Red	Intermediate	White

Skeletal muscle fibers (cells) can be identified as types 1 (slow oxidative), 2A (fast oxidative-glycolytic), or 2X (fast glycolytic); this classification scheme is based on two features of muscle fibers:

- Their **contractile speed** (fast or slow). Contractile speed is determined largely by the catalytic speed of the **myosin ATPase** and the activity of the associated motor neuron at the **neuromuscular junction**.
- Their **primary means of producing ATP** (oxidative or glycolytic). Oxidative fibers are specialized to use **aerobic respiration**, whereas glycolytic fibers are specialized to use **anaerobic glycolysis**.

Type 1 fibers are fatigue resistant, contract slowly, and use aerobic respiration as their primary method of ATP production. Consequently, compared with glycolytic fibers, type 1 fibers are much **richer in mitochondria (Number I)**, the site where many enzymes involved in aerobic respiration reside. Type 1 fibers also have access to a continuous supply of oxygen (the final electron acceptor of aerobic respiration) through an **extensive network of capillaries (Number II)** that surround the muscle fiber. In addition, type 1 fibers appear red under the microscope due to **high concentrations of myoglobin**, a red-pigmented, oxygen-binding molecule that aids in delivery of oxygen to the muscle cell interior (**Number III**).

Educational objective:

Skeletal muscle fibers can be classified as type 1 (slow oxidative), type 2A (fast oxidative-glycolytic), or type 2X (fast glycolytic). The speed of contraction (slow or fast) is determined by the catalytic speed of the myosin ATPase and the activity of the fiber's associated motor neuron. Oxidative fibers are fatigue resistant and specialized for synthesis of ATP through aerobic respiration (oxygen-requiring pathways). Glycolytic fibers are easily fatigable and specialized to generate ATP through anaerobic glycolysis (which does not require oxygen).

Time spent:0 sec

Subject:
Biology

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System:
Musculoskeletal System